

Multi-Agent Systems: Patterns & Orchestration

Frontend
Backend
MLE
QA
DS/DA



Human
LLMs

"context"

Single Agent → simple / standalone task

coherent

few tools

no need of specialization

low latency

easier debugging

} Single Agent

System prompt is large / context is very large

Too much boiler

Task has diff phases

planning
research
coding
writing

Multiple workstreams can run parallelly

Need of multi-agent

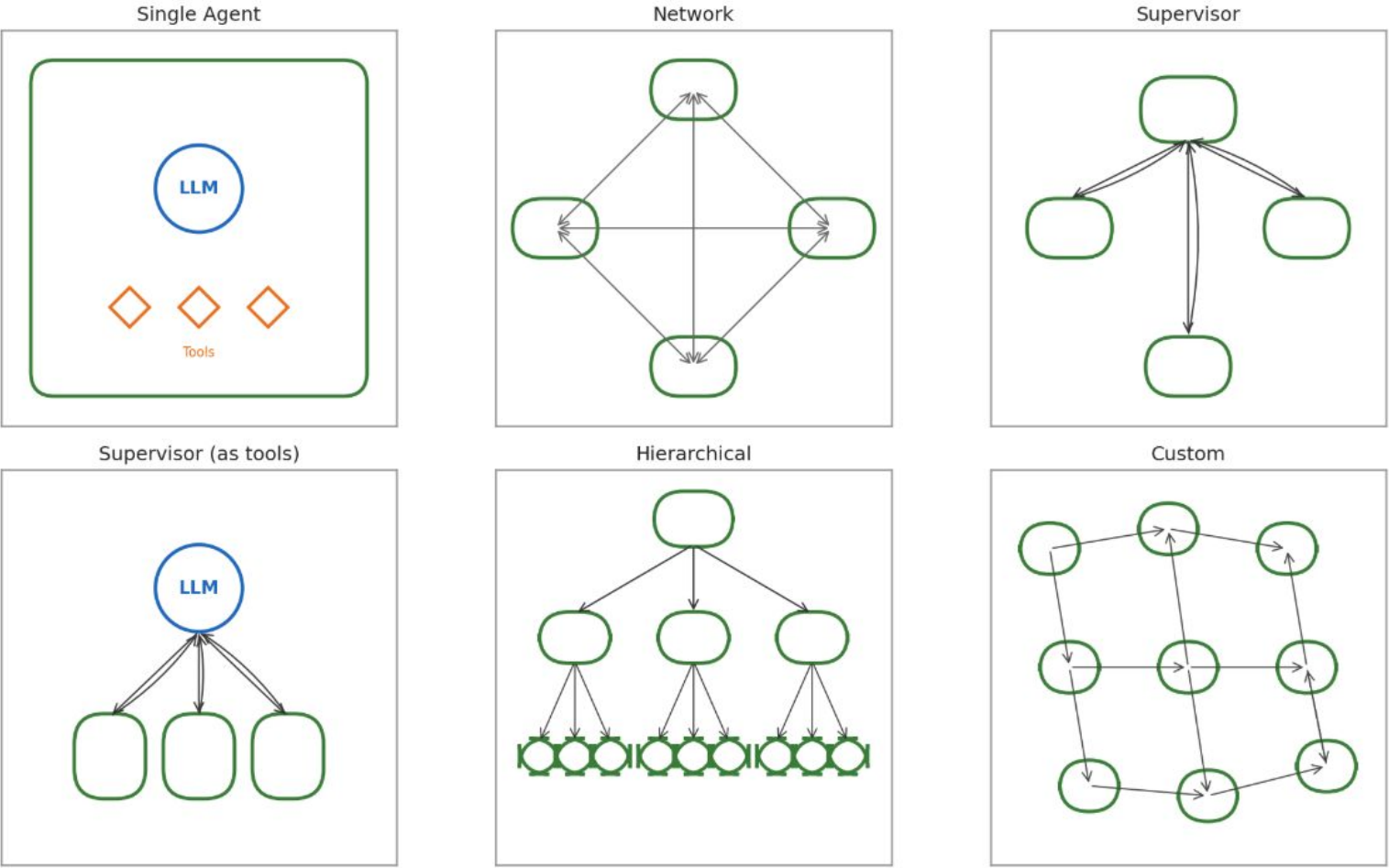
~~Single Agent~~

~~Context bloat~~

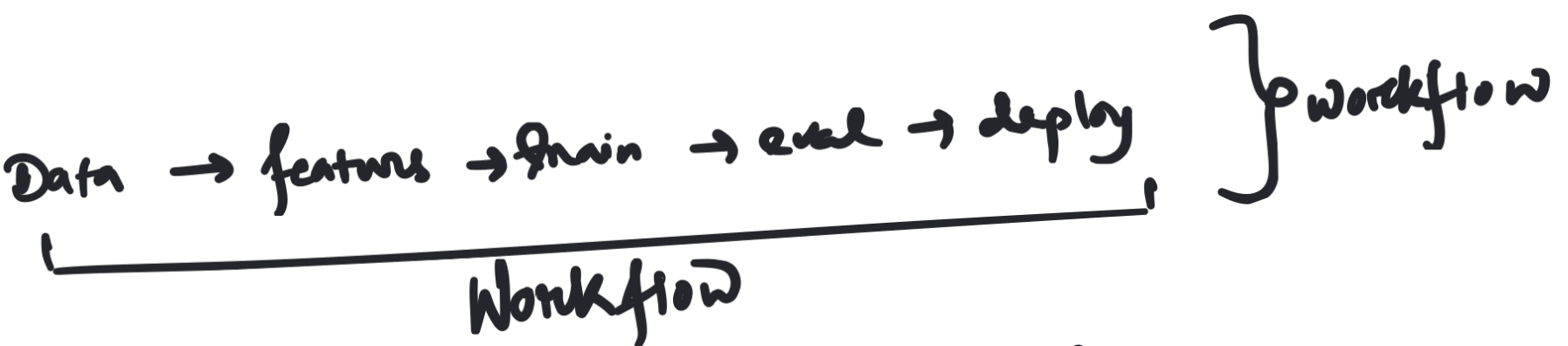
~~Role confusion~~

~~no specialization~~

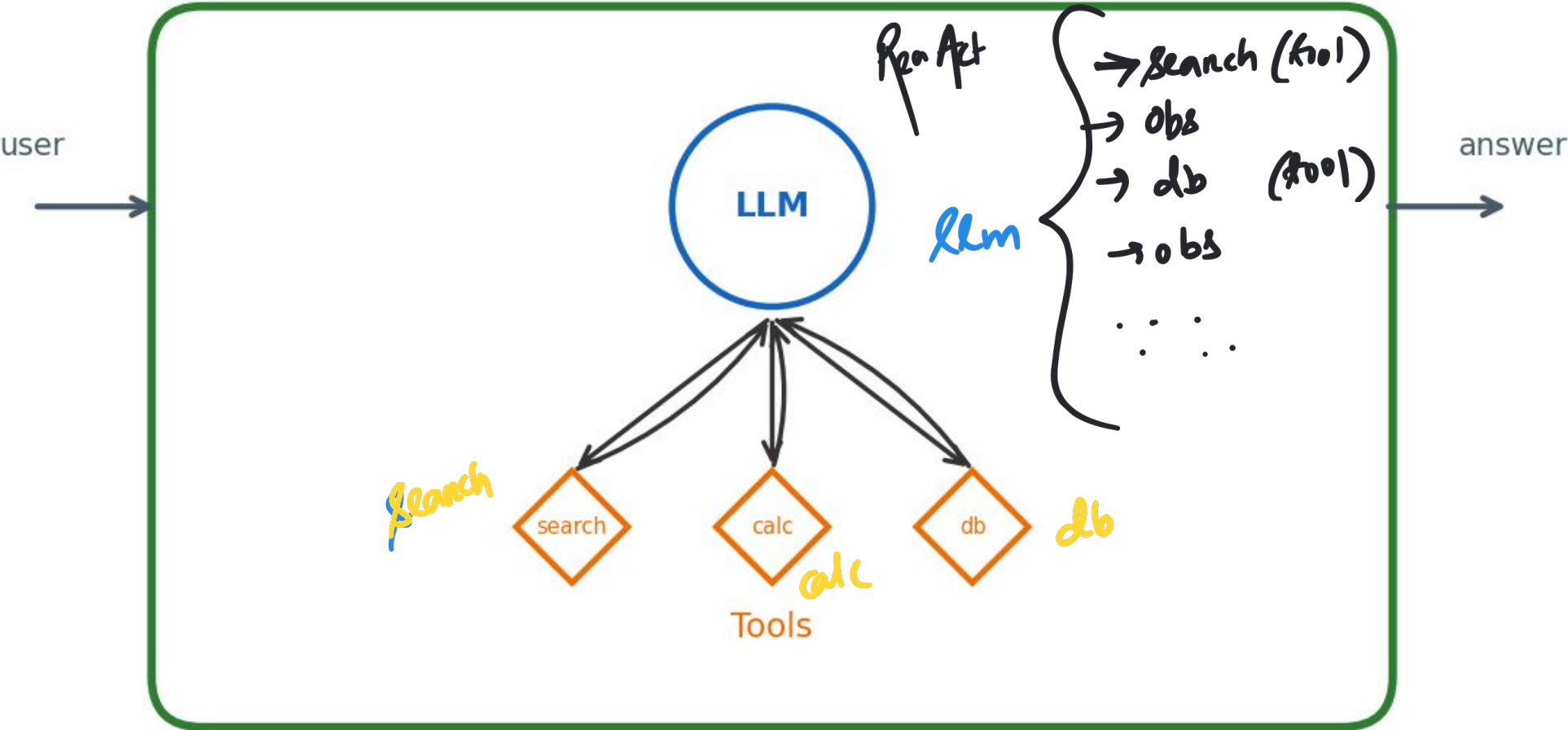
Six common multi-agent architecture patterns

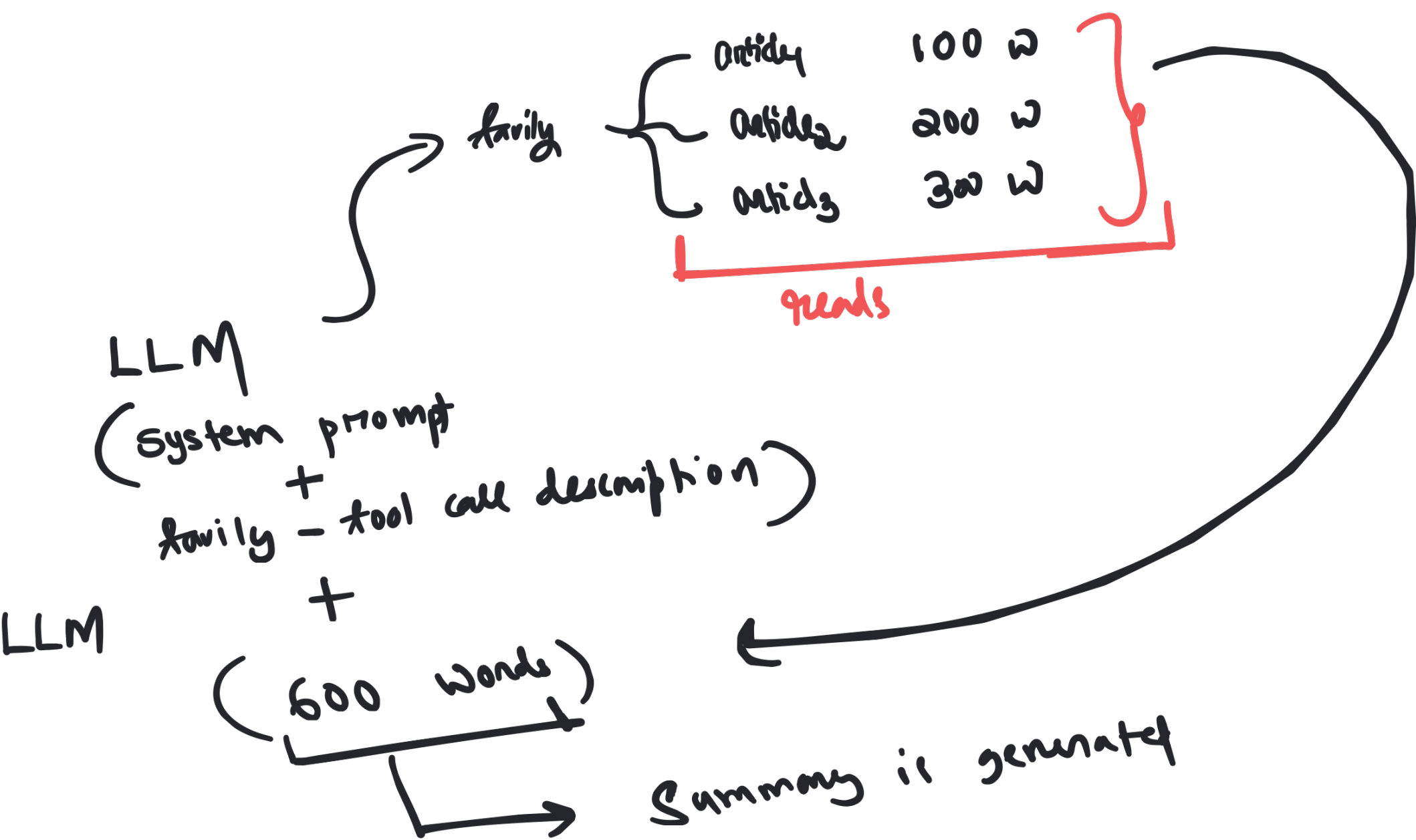


Six common multi-agent architecture patterns. Single Agent (top-left) is the baseline; everything else trades simplicity for some combination of specialization, autonomy, or scale.

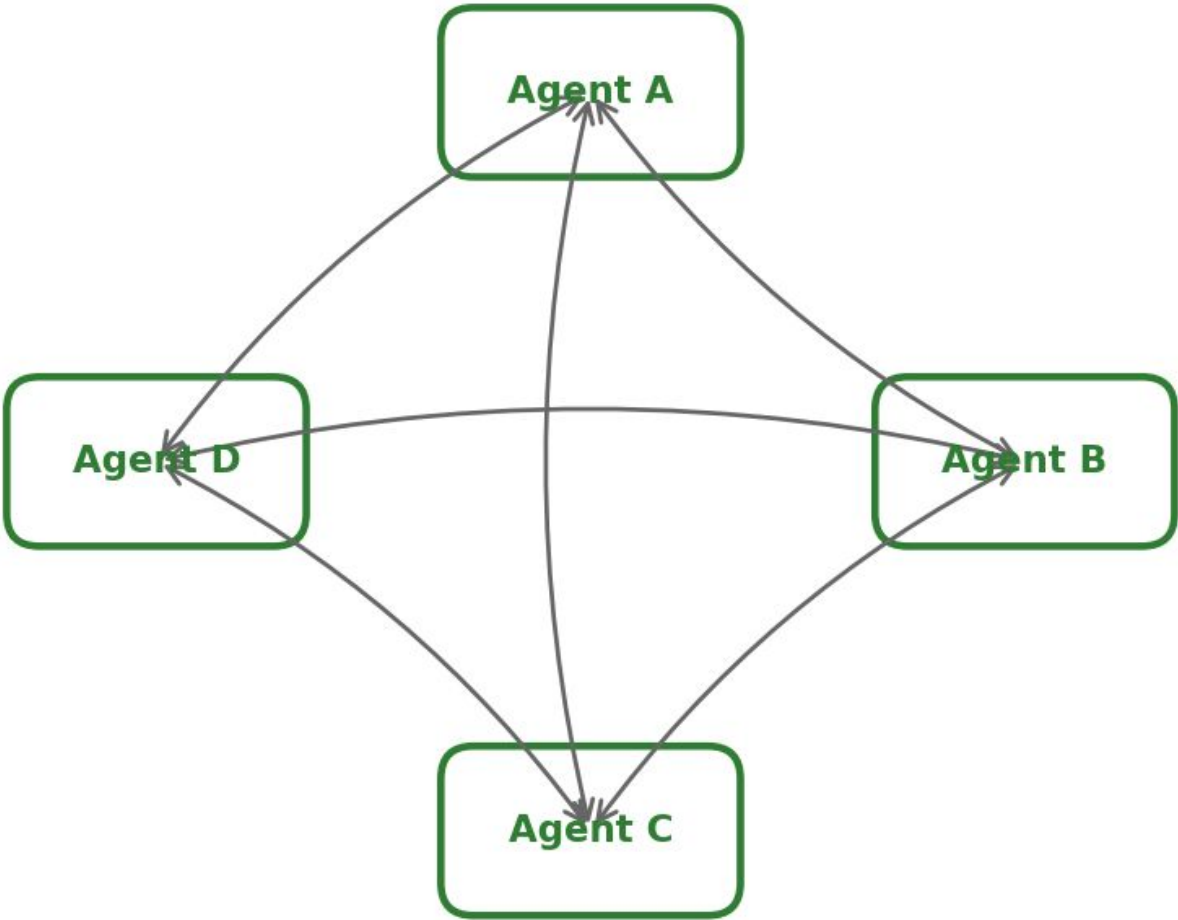


Single Agent with Tools

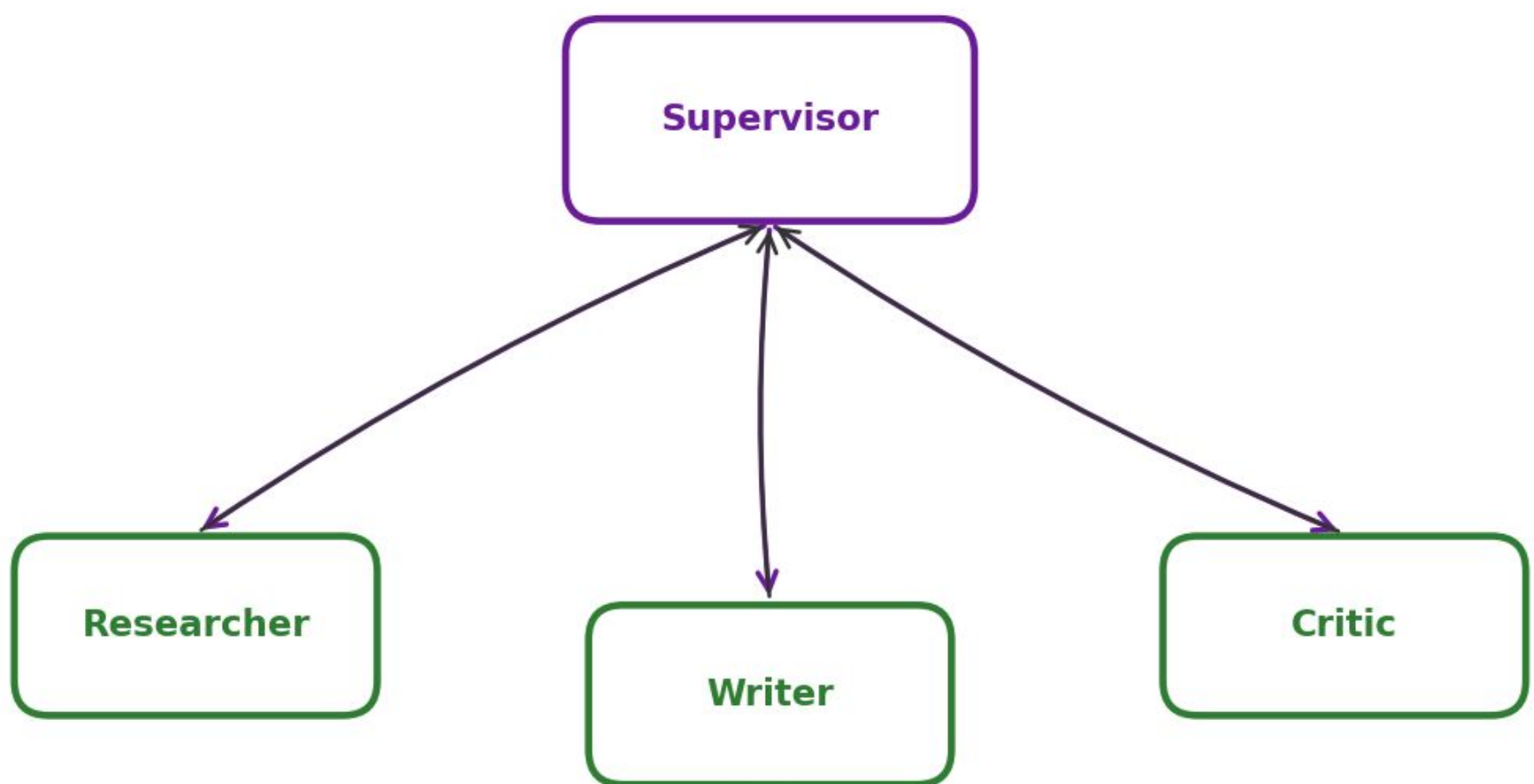




Network Pattern — agents talk to each other freely



Supervisor Pattern — central router delegates to specialists



UQ

(delegate
correctly)

Supervisor

—

Agent 1

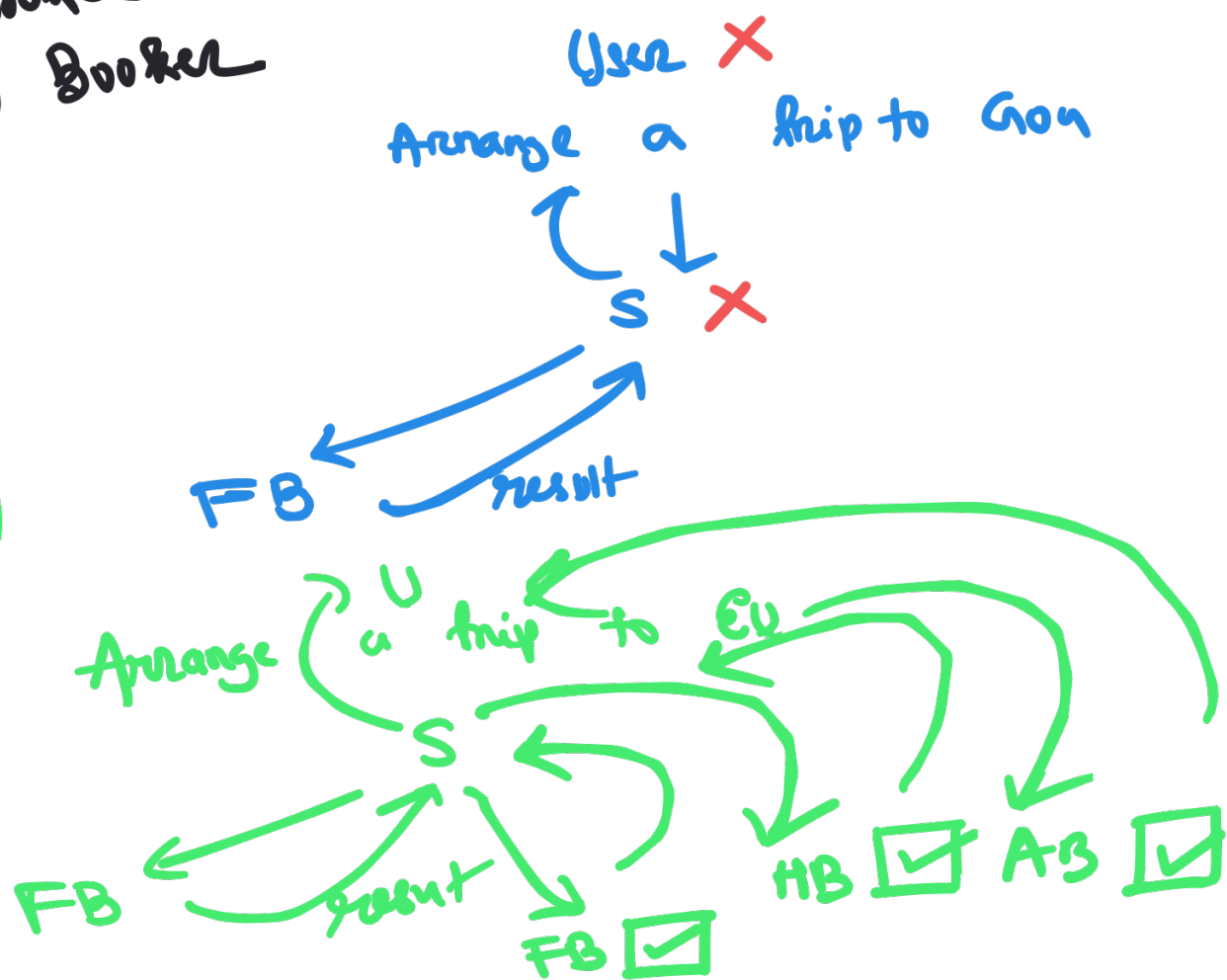
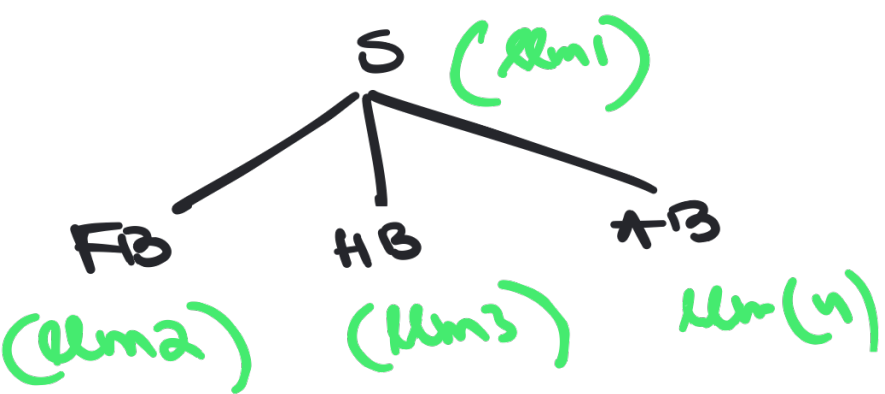


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Agent 3



Agent1 → Flight Booker
 Agent2 → Hotel Booker
 Agent3 → Activity Booker

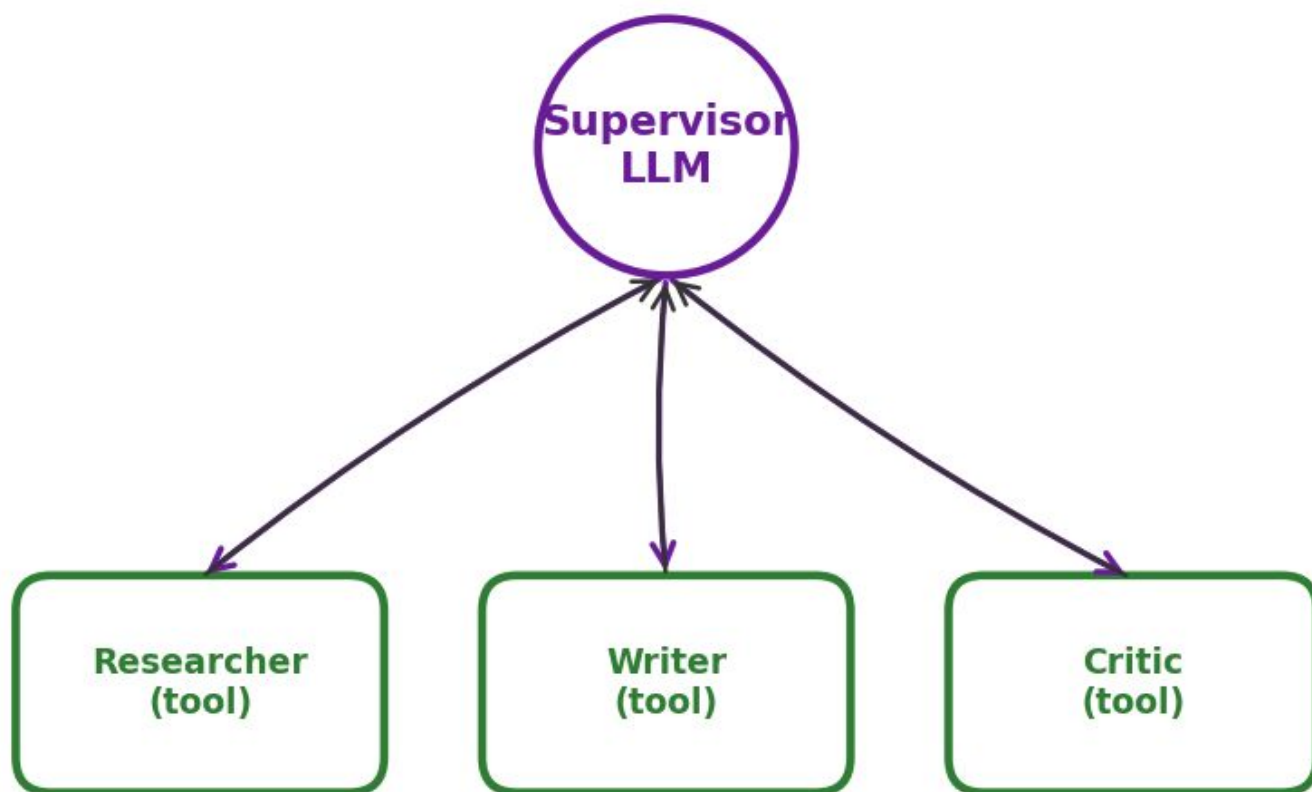


Having different Lms for
Each agent

4 diff prompts $\Rightarrow$ Context Management
is good ✓

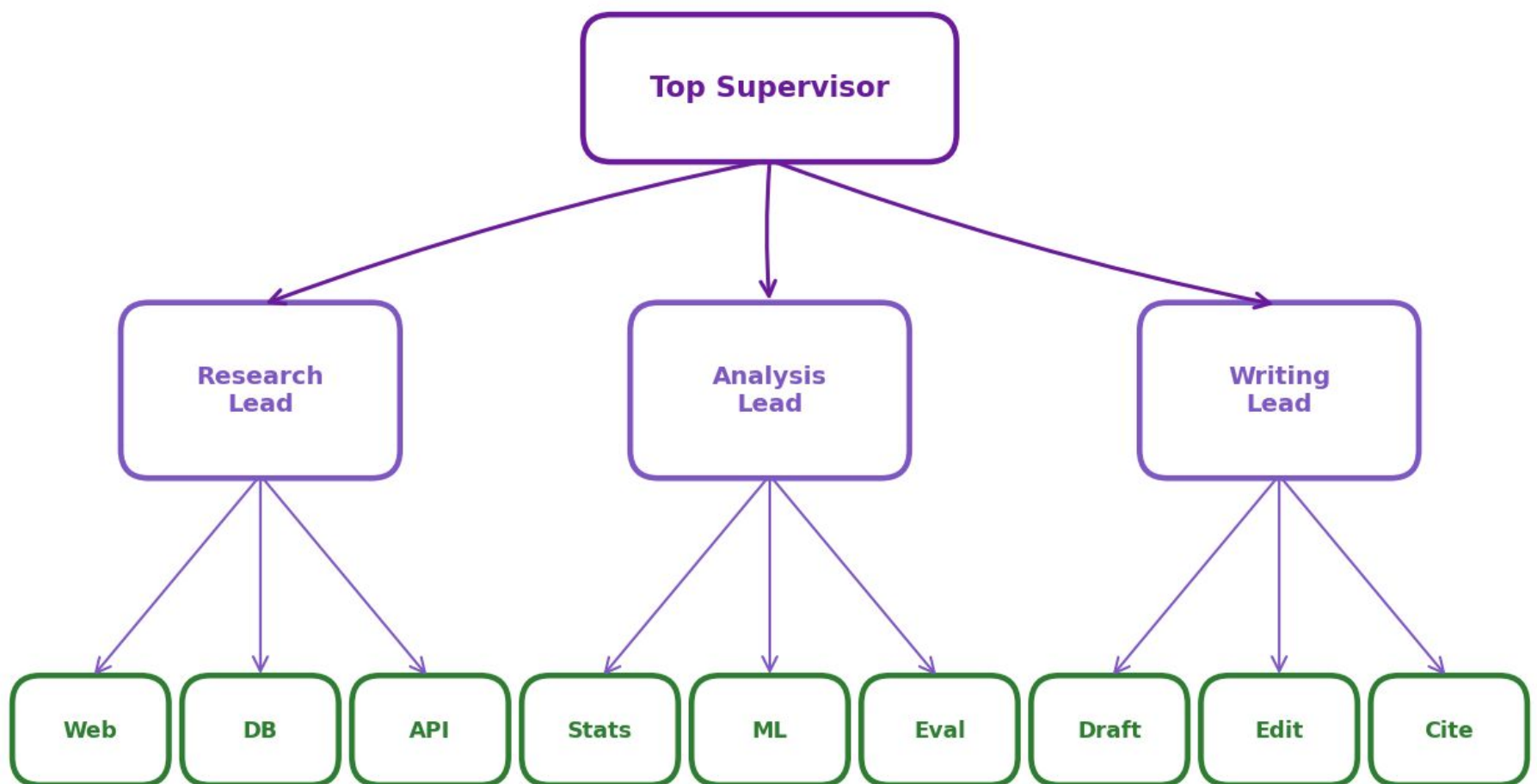
- $\rightarrow$ FB
- $\rightarrow$ AG
- $\rightarrow$ HB

Supervisor-as-Tools — agents wrapped as tool calls



Specialist agents are exposed as callable tools

Hierarchical Pattern — supervisors of supervisors



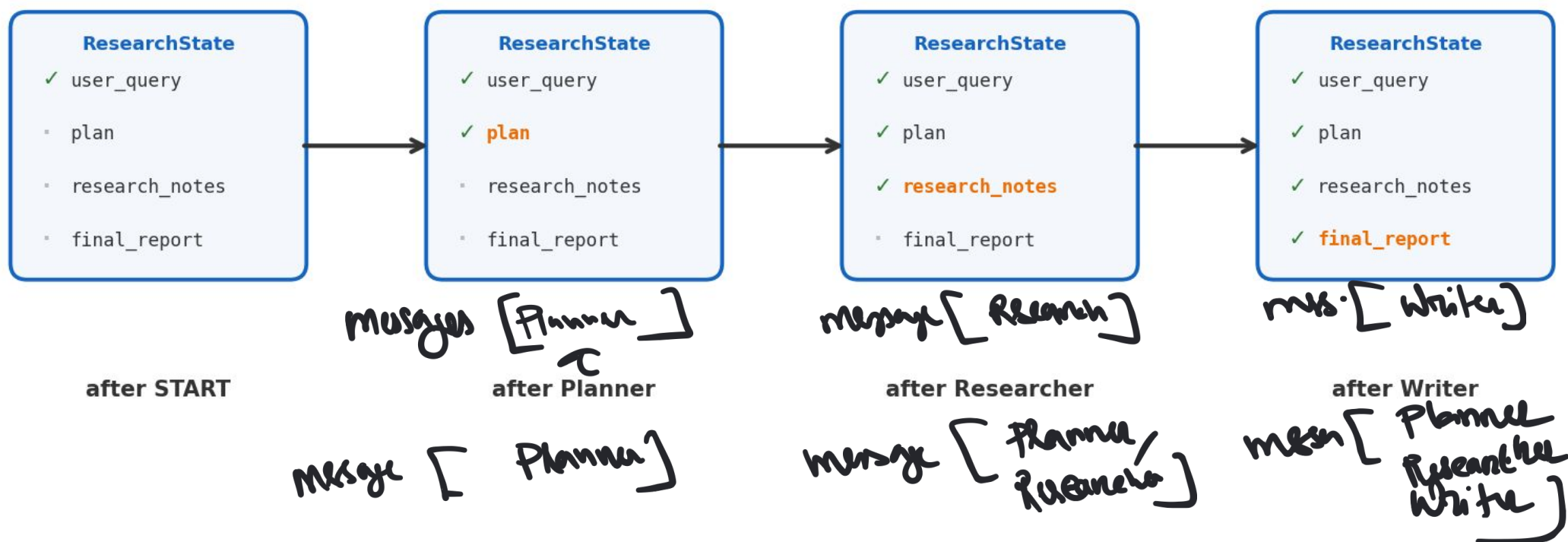
Sequential Research Team — state accumulates as it flows through nodes



x Messages : Annotated [List [DequeMessage], add_message] ✓✓

State Flow — each node reads from and writes to the shared state

Deterministic Orchestration



- ✓ $\text{foo} \rightarrow \text{add}$
- ✓ $\text{foo} \rightarrow \text{mult}$
- ✓ $\text{foo} \rightarrow \text{div} \times$
- ✓ $\text{foo} \rightarrow \text{sub} \times$

$$\begin{array}{c} \text{OR} \\ \hline 9 * 5 + 5 \\ \hline \end{array}$$

Math
LLM

rule - BODMAS

$\text{foo} \rightarrow \text{mult}(9, 5)$
 result
 $\text{foo} \rightarrow \text{add}(\text{result}, 5)$
 final result

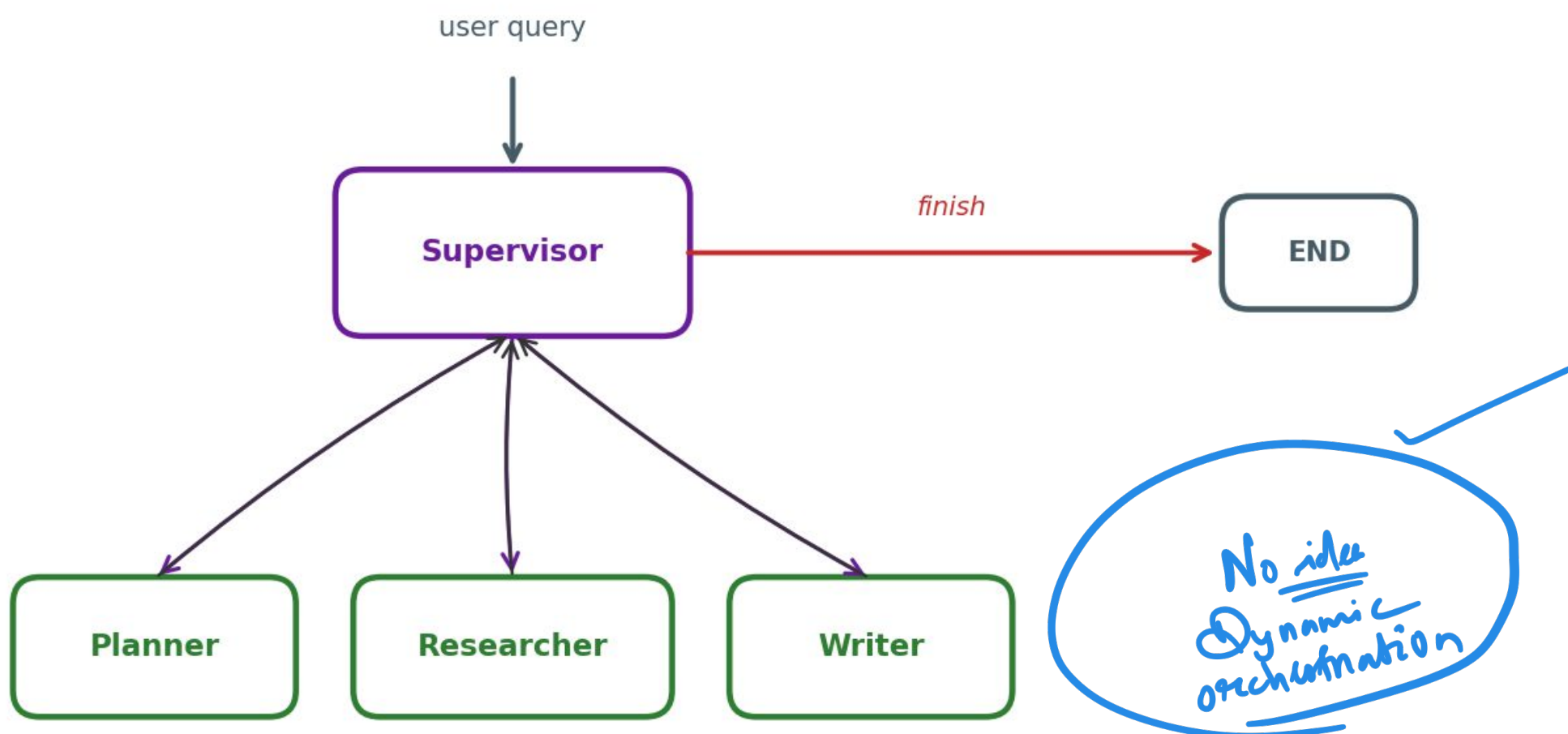
✓ Supervisor

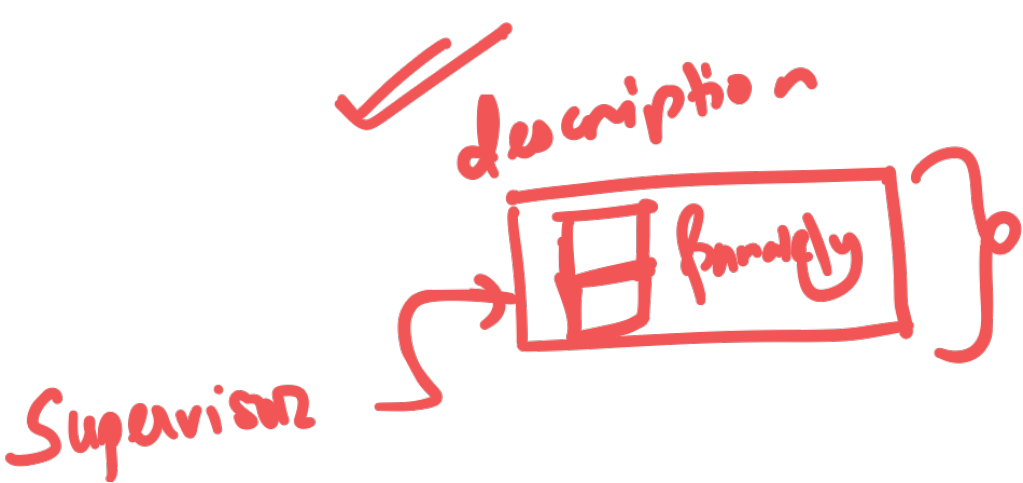
✓ Subagent 1
description ✓

Subagent 2 ✓
description ✓

Tools for subagent → won't be known to supervisor

Supervisor-Controlled Team — supervisor decides who runs next (and when to finish)





Supervisor ends — [Max supervisor limit

End (final report)
in
this case

Supervisor

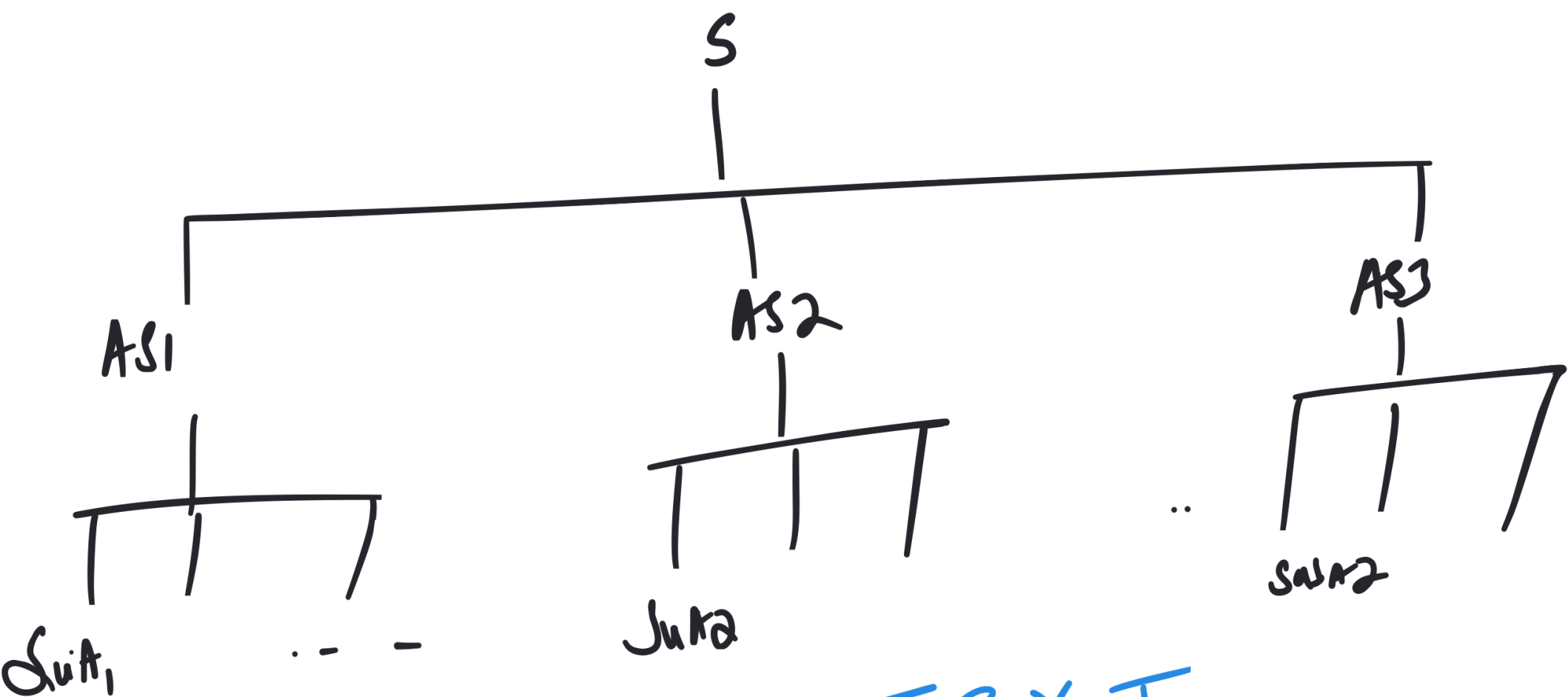
S1 - A1

S2 - K2

S3 - K3

S4 - A4

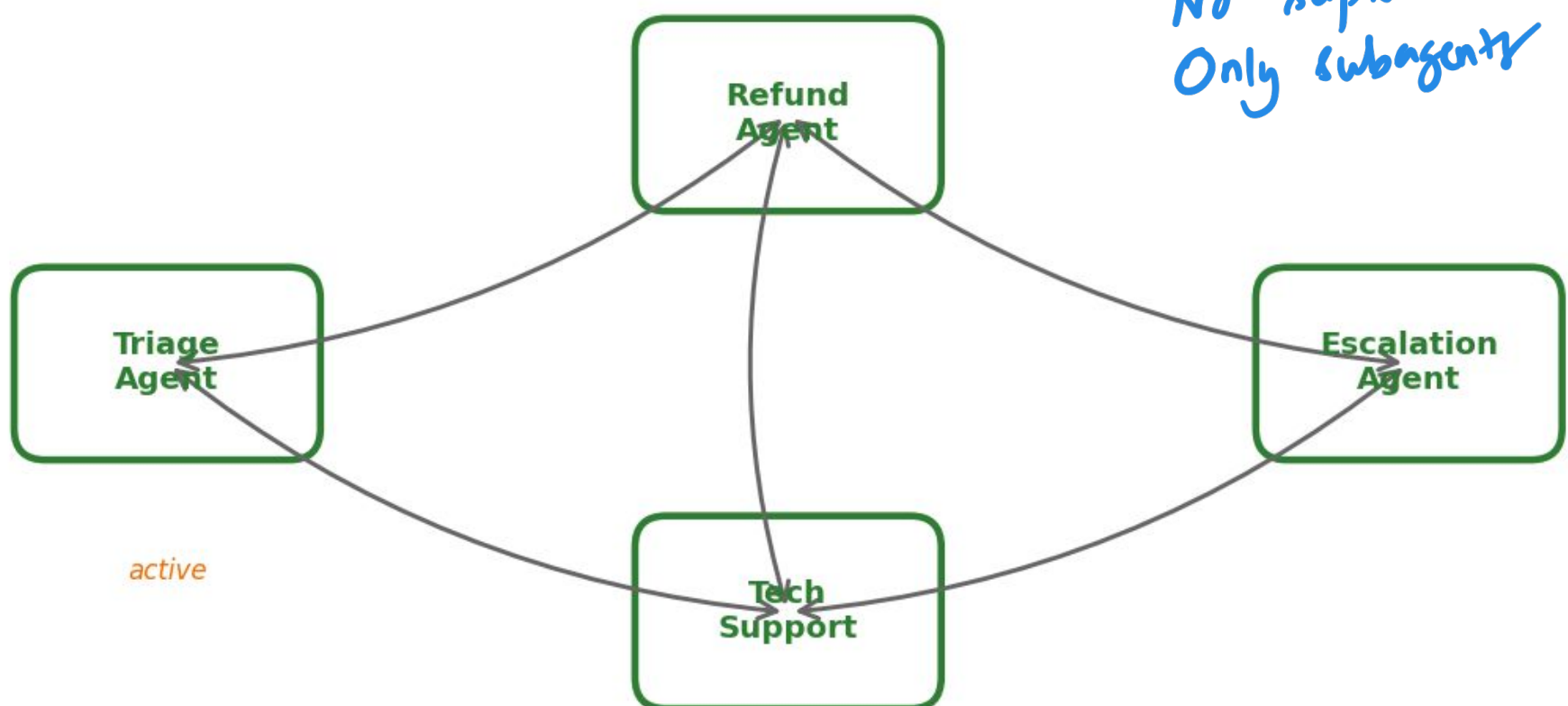




CONTEXT
(of LLM)

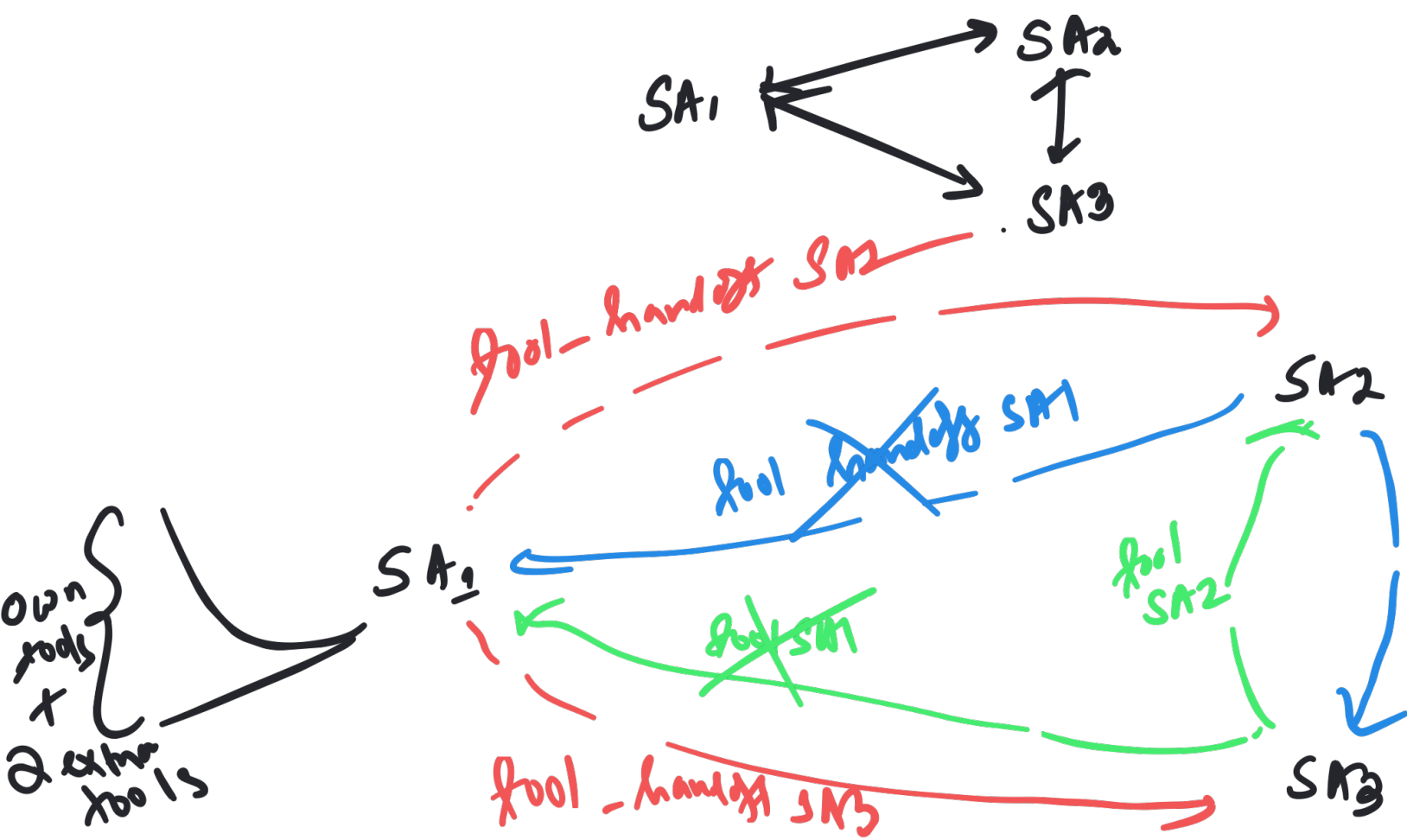
Handoff / Swarm — control transfers between peers

*No supervisor
Only subagents*



Each agent can hand off control to another via a "transfer_to_X" tool

Side
tool for handoff



Flight ○
Hotel ○

